INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

Lab 3: Advanced GRC Phishing Simulation & Quantitative Risk Analysis

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PHASE 1: BASIC METRIC CALCULATIONS

Step 1: Calculate Fundamental Rates

Campaign	Email Open Rate	Click-Through Rate	Credential Submission Rate	Data Entry Rate
A	(82/100)×100 = **82%**	(47/100)×100 = **47%**	(18/100)×100 = **18%**	(5/100)×100 = **5%**
B	(200/250)×100 = **80%**	(150/250)×100 = **60%**	(75/250)×100 = **30%**	(25/250)×100 = **10%**
C	(72/80)×100 = **90%**	(60/80)×100 = **75%**	(12/80)×100 = **15%**	(8/80)×100 = **10%**

Step 2: Calculate Efficiency Metrics

How effective is each step of the attack?

Conversion Rate from Open to Click:

- Campaign A: (47/82)×100 = 57.3%
- Campaign B: (150/200)×100 = 75%
- Campaign C: (60/72)×100 = 83.3%

Post-Click Credential Submission Rate:

- Campaign A: (18/47)×100 = 38.3%
- Campaign B: (75/150)×100 = 50%
- Campaign C: (12/60)×100 = 20%

PHASE 2: ADVANCED RISK QUANTIFICATION

Step 3: Calculate Organizational Risk Exposure

For Campaign B (most successful):

- Expected Credentials Stolen: 2,000 × (75/250) = 600 employees
- Expected Data Breaches: 600 × 25% probability = 150 breaches

CALCULATION:

- Campaign B Credential Rate: 75/250 = 30%
- Organization Exposure: 2,000 × 0.30 = 600 employees
- Expected Breaches: 600 × 0.25 = 150 breach incidents

Step 4: Financial Impact Analysis

A. Calculate Potential Financial Loss:

- Single Breach Cost: $4,350,000
- Expected Number of Breaches: 150
- Total Exposure: $4,350,000 × 150 = $652,500,000

B. Calculate Annualized Loss Expectancy (ALE):

ALE = Single Loss Expectancy × Annual Rate of Occurrence

- Single Loss Expectancy (SLE): $4,350,000
- Annual Rate of Occurrence (ARO): 150 expected breaches
- ALE = $4,350,000 × 150 = $652,500,000

B. Calculate Annualized Loss Expectancy (ALE):

ALE = Single Loss Expectancy × Annual Rate of Occurrence

- Single Loss Expectancy (SLE): $4,350,000
- Annual Rate of Occurrence (ARO): 150 expected breaches
- ALE = $4,350,000 × 150 = $652,500,000

Step 5: Control Effectiveness Analysis

MFA Cost-Benefit Analysis:

- Cost to Implement MFA: $45 per user × 2,000 users = $90,000
- MFA Effectiveness: 99.9% reduction in credential theft impact
- Risk Reduction: $652,500,000 × 0.999 = $651,847,500

- ROI: ($651,847,500 - $90,000) / $90,000 = 724,175%

Security Training Cost-Benefit:

- Training Cost: $50 per user × 2,000 users = $100,000
- Expected Effectiveness: 60% reduction in click-through rates
- New Click-Through Rate: 30% × (1-0.60) = 12%
- New Credentials Stolen: 2,000 × 0.12 = 240
- New Expected Breaches: 240 × 0.25 = 60
- New ALE: $4,350,000 × 60 = $261,000,000
- Risk Reduction: $652,500,000 - $261,000,000 = $391,500,000
- ROI: ($391,500,000 - $100,000) / $100,000 = 391,400%

PHASE 3: COMPREHENSIVE RISK ASSESSMENT

Step 6: Complete Risk Assessment Matrix

Campaign	Overall Risk Score	Financial Exposure	Priority Level	Recommended Action
A	MEDIUM-HIGH	$156,600,000	2	Enhanced Training
B	**CRITICAL**	**$652,500,000**	**1**	**IMMEDIATE MFA**
C	HIGH	$104,400,000	2	Executive Training

CALCULATIONS FOR TABLE:

- Campaign A Exposure: (18/100)×2000 = 360 employees × 0.25 × $4,350,000 = $156,600,000
- Campaign C Exposure: (12/80)×2000 = 300 employees × 0.25 × $4,350,000 = $104,400,000

Step 7: Statistical Significance Analysis

Calculate Confidence Intervals (95% Confidence):

Formula: $p \pm 1.96 \times \sqrt{[p(1-p)/n]}$*

For Campaign B Credential Rate (30%):

- $p = 0.30, n = 250$
- Margin of Error = $1.96 \times \sqrt{[0.30(1-0.30)/250]} = 1.96 \times \sqrt{(0.21/250)} = 1.96 \times \sqrt{0.00084} = 1.96 \times 0.029 = 0.057$
- True Rate Range: 24.3% to 35.7%

Calculate for Campaign A: Credential Rate : 18%

- $p = 0.18, n = 100$
- Margin of Error = $1.96 \times \sqrt{[0.18(1-0.18)/100]} = 1.96$ x $0.0384 = 0.0753 = 7.53\%$
- True Rate Range: $18\% \pm 7.53\% = 10.47\%$ to 25.53%

PHASE 4: EXECUTIVE REPORTING

Step 8: Comprehensive Executive Briefing

To: Board of Directors, CISO, CFO

From: GRC Risk Analysis Team

Date: 31ST October,2026

Subject: CRITICAL: Quantitative Phishing Risk Assessment & $652M Exposure

1. Executive Summary:

Our phishing simulation analysis reveals a CRITICAL risk exposure of $652 million annually. The "IT Password Reset" campaign showed 30% credential theft rate, projecting to 600 compromised accounts organization-wide. Immediate action is required.

2. Key Quantitative Findings:

Risk Metric	Value	Industry Average	Severity
Overall Credential Theft Rate	22.1%	15.3%	HIGH

Maximum Campaign Success Rate	30%	18.7%	CRITICAL
Annualized Loss Expectancy	$652M	$285M	SEVERE
MFA Implementation ROI	724,175%	350%	EXCELLENT

3. Detailed Financial Analysis:

Current State Risk:

- Annualized Loss Expectancy: $652,500,000
- Probability of Major Breach: 93.75% (150 expected breaches)
- Per-Employee Risk: $326,250

Proposed Security Investment Portfolio:

Control	Cost	Risk Reduction	Net Benefit	ROI	Priority
MFA Implementation	$90,000	$651.8M	$651.7M	724,175%	**1**
Enhanced Training	$100,000	$391.5M	$391.4M	391,400%	**2**
Email Filtering	$50,000	$130.5M	$130.4M	260,800%	**3**

4. Strategic Recommendations:

IMMEDIATE ACTIONS (Q1):

1. Implement MFA enterprise-wide - $90,000 investment
2. Launch targeted phishing simulation program - $25,000
3. Conduct emergency security awareness training - $100,000

STRATEGIC INITIATIVES (FY2024):

1. Deploy advanced email security gateway - $50,000
2. Establish continuous security monitoring - $150,000
3. Implement security behavior analytics - $75,000

5. Expected Risk Posture After Controls:

Metric	Current	With Controls	Reduction
Credential Theft Rate	22.1%	2.2%	90%
Annual Loss Expectancy	$652M	$65M	90%
Breach Probability	93.75%	25%	73.3%

Total Investment Required: $490,000

Total Risk Reduction: $587,500,000

Net Financial Benefit: $587,010,000

BONUS CHALLENGE:

Calculate the break-even point for our security investments. If we invest $490,000 in controls, how many breaches must we prevent to justify the cost?

Hint: $490,000 / $4,350,000 per breach = 0.11breaches